

Operating Instructions

Diesel engine

12 V 4000 S83

16 V 4000 S83

MS150026/02E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
12V4000S83	139.8 kW/cyl.	4D, Frac operation (continuous operation at low load)
16V4000S83	139.8 kW/cyl.	4D, Frac operation (continuous operation at low load)

Table 1: Applicability

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Safety

1.1 Important provisions for all products

Nameplate

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the safety instructions and warning notices

Correct use

The product is intended exclusively for the application specified in the contract or defined at the time of delivery.

This means that the equipment must be operated:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine control/parameters)
- In compliance with all safety regulations and in adherence with all warning notices in this manual
- In accordance with the maintenance requirements over the entire service life of the product (→ Maintenance Schedule)
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use is considered improper use and increases the risk of personnel injury or material damage in product operation. The manufacturer will accept no liability for such damage.

Modifications or conversions

Unauthorized modifications to the product compromise safety.

The manufacturer will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine spare parts must be used to replace components or assemblies.

The manufacturer will accept no liability or warranty claims for any damage caused by the use of other spare parts.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

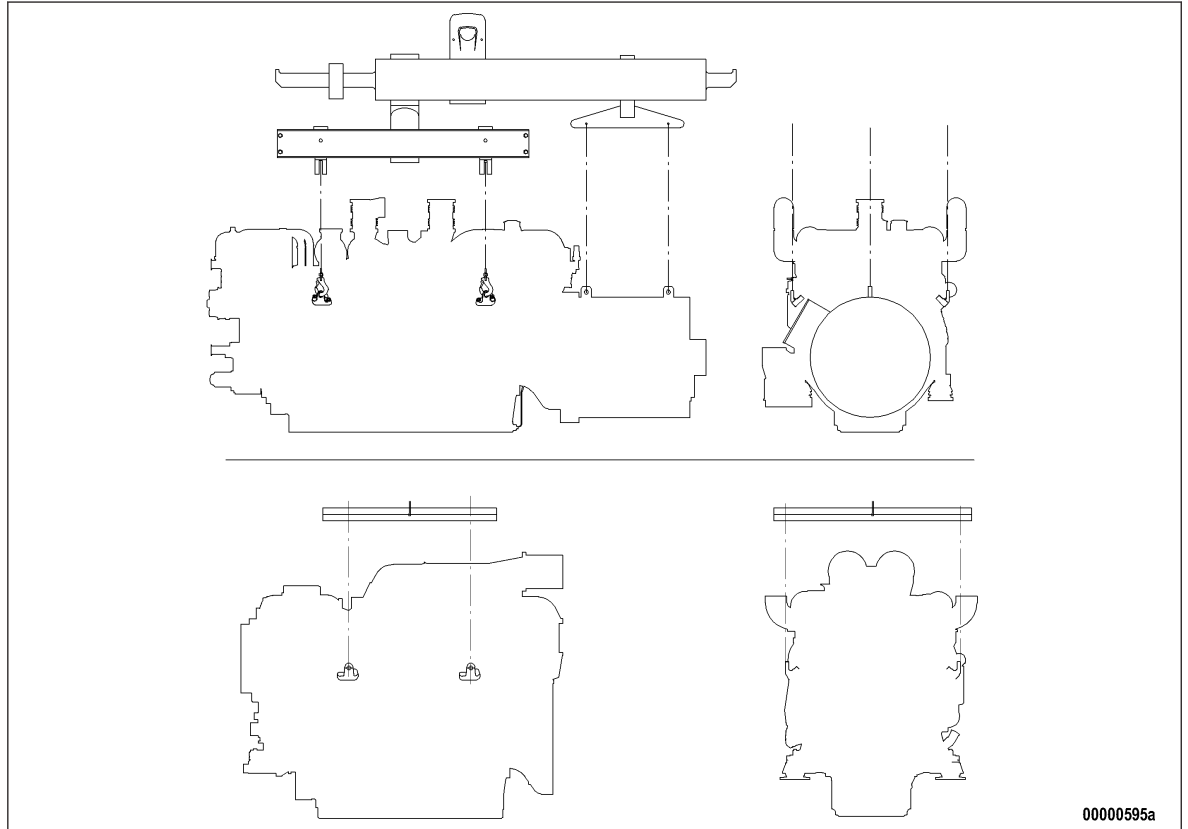
Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Transport

Transport



Always use the lifting eyes on the engine and generator/gearbox when transporting gensets.

Always use the lifting eyes on the engine when transporting an engine separately.

Only use transport and lifting devices approved by MTU.

The engine/genset must only be transported in installation position: maximum permissible diagonal pull 10°.

Remove any loose parts on the genset.

Raise the engine/genset slowly. The lifting ropes or chains must not make contact with the engine or its components. Readjust lifting device as necessary.

Pay attention to the center of gravity of the engine/genset.

For special packaging with aluminium foil: Suspend the engine/genset by the lifting eyes on the bearing pedestal or transport by means of handling equipment (forklift truck) capable of bearing the load.

Fit the crankshaft transportation lock on the engine and fit the engine mount locking devices prior to transport.

Secure the engine/genset such as to preclude tipping during transport. Secure such as to preclude slipping and tipping when driving up or down inclines and ramps.

Placement after transport

Place the engine/genset on a firm, flat surface only.

Make sure that the consistency and load-bearing capacity of the ground or support surface is adequate.

Never set an engine down on the oil pan unless expressly authorized to do so by MTU on a case-to-case basis.

1.4 Crankshaft transport locking device

Special tools, Material, Spare parts

Designation / Use	Part No.	Qty.
Torque wrench, 10-60 Nm	F30452769	1
Torque wrench, 60-320 Nm	F30452768	1
Engine oil		

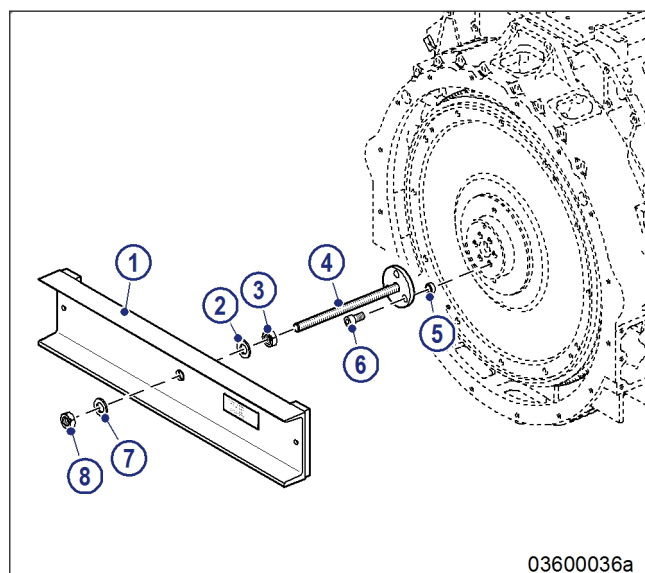
Note: The locking device protects the crankshaft bearings from shocks and vibration damage during engine transport.

For installation and removal of the transport locking device, follow the instructions below:

1. The transport locking device must remain installed as long as possible during engine installation in order to avoid damage.
2. Prior to every engine transport, the transport locking device must be reinstalled on both sides according to the instructions.
3. If the engine is to be moved together with the generator, the transport locking device for the generator must also be installed.
4. Always use the screws supplied with or installed in the transport locking device to secure it on the engine.
5. Starting or barring the engine is allowed only with the transport locking device removed. If the generator/transmission is already mounted on the engine, ensure that the transport locking device of the generator/transmission is also removed.

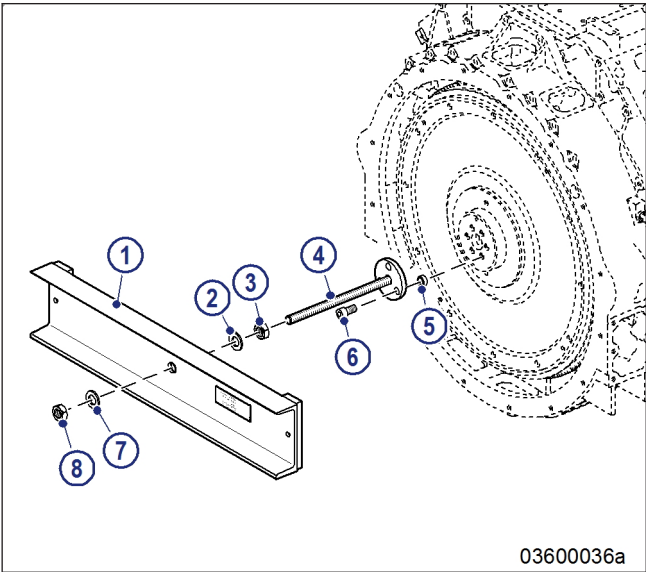
Removing the transport locking device from driving end (KS), variant A

1. Remove parts (1) to (8) from engine. Spacer sleeves (5) are optional.
2. Store the removed parts of the transport locking device carefully for possible reuse.



Installing the transport locking device on driving end (KS), variant A

1. Screw retainer (4) with spacer sleeves (5) and screws (6) on crankshaft. Spacer sleeves (5) are optional.



2. Use torque wrench to tighten screws (6) to specified tightening torque .

Name	Size	Type	Lubricant	Value/Standard
Screw	M16	Tightening torque	(Engine oil)	160 Nm + 16 Nm

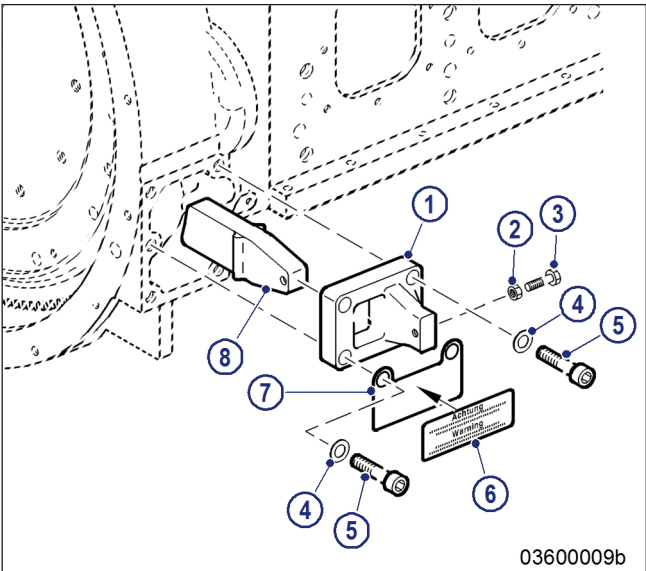
3. Screw nut (3) with washer (2) onto retainer (4). Fit support (1) with washer (7) and nut (8) and tighten nut (8) with torque wrench to the specified tightening torque.

Name	Size	Type	Lubricant	Value/Standard
Nut	M24	Tightening torque	(Engine oil)	100 Nm + 10 Nm

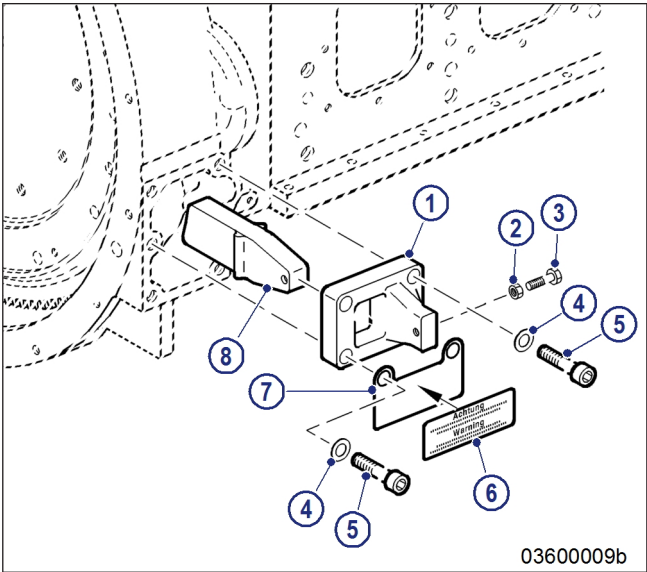
4. Secure with nut (3).
5. Mark the engine as "Fitted with transportation locking device".

Removing the transport locking device from driving end (KS), variant B

1. Release the locknuts (2) on both sides of the flywheel housing, remove screw (3) and take off retainer (8).
2. Remove screws (5) and take off with washers (4), cover plate (7), label (6) and plates (1).
3. Store the removed parts of the transport locking device together with this documentation carefully for possible reuse.



Installing the transport locking device on driving end (KS), variant B



- 1. Secure plate (1) with screws (5) and washers (4) together with cover sheet (7) at the bores on both sides of the flywheel housing and tighten to the specified tightening torque.

Name	Size	Type	Lubricant	Value/Standard
Screw	M16	Tightening torque	(Engine oil)	250 Nm +25 Nm

- 2. Screw locknut (2) onto screws (3) up to the end of the thread.
- 3. Fit the retainers (8) through the openings of plate (1) and fasten with screw (3).
- 4. Tighten screws (3) alternately with torque wrench to the specified tightening torque.

Name	Size	Type	Lubricant	Value/Standard
Screw	M10	Tightening torque	(Engine oil)	30 Nm +3 Nm

- 5. Place locknut (2) of screw (3) in position on plate (1) and lock it.
- 6. Fit label (6) to mark the engine as "Fitted with transportation locking device".